```
r1.low, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.276             -37568.7            -37550.7                 0
     2 20263017             5000           0.276             -37553.4            -37544.4                 0
     3 20263018             5000           0.275             -37570.0            -37545.4                 0
     4 20263019             5000           0.273             -37560.1            -37543.4                 0
                                    file
 mcmc_r1.low_step2_chain1_2026-10-30.Rdata
 mcmc_r1.low_step2_chain2_2026-10-30.Rdata
 mcmc_r1.low_step2_chain3_2026-10-30.Rdata
 mcmc_r1.low_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
                                      trate.0                       female.to.male.multiplier
                                        2.112                                           1.898
          age.5.to.14.hiv.mortality.multiplier.0          age.20.to.29.transmission.multiplier.3
                                        1.823                                           1.552
          age.20.to.29.transmission.multiplier.0                male.hiv.mortality.multiplier.0
                                        1.505                                           1.504
          age.40.to.49.transmission.multiplier.0          age.15.to.19.transmission.multiplier.0
                                        1.416                                           1.375
          age.30.to.39.transmission.multiplier.3    age.45.to.64.mortality.slope.multiplier.male
                                        1.312                                           1.310
          age.15.to.19.transmission.multiplier.3                male.hiv.mortality.multiplier.2
                                        1.306                                           1.298
age.65.to.79.mortality.intercept.multiplier.male                                 age.assortativity
                                        1.286                                           1.280
                                      trate.4                 hiv.specific.mortality.rates.2
                                        1.279                                           1.263
       over.80.mortality.slope.multiplier.female         age.15.to.24.hiv.mortality.multiplier.0
                                        1.256                                           1.246
          age.20.to.29.transmission.multiplier.1                                       trate.3
                                        1.236                                           1.229
age.45.to.64.mortality.intercept.multiplier.male    age.65.to.79.mortality.slope.multiplier.male
                                        1.226                                           1.208
  age.65.to.79.mortality.slope.multiplier.female         age.25.to.49.hiv.mortality.multiplier.0
                                        1.189                                           1.187
          age.30.to.39.transmission.multiplier.1
                                        1.186

Merged simset:
 location simulations total.weight                                          file
   r1.low        1000         1000 simset_r1.low_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 16 of 33 targets flagged
     data.type         age  sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality       25-49 both              26                   0              x0.36      2025            x0.31
 hiv.mortality 50 and over both              26                   8              x1.95      2025            x1.56
    engagement        0-14 both              11                   0          -16.2 pts      2025         +1.6 pts
 hiv.mortality       10-19 both              26                  23              x1.34      2025            x1.46
   suppression        0-14 both              11                  18          -12.3 pts      2025         +2.2 pts
     awareness        0-14 both              11                   9           -4.9 pts      2025         -7.3 pts
     awareness         15+ both              11                   0           +3.6 pts      2025         +2.8 pts
 hiv.mortality        0-14 both              26                  42              x1.17      2025            x1.03
     incidence       15-24 both              26                  38              x0.88      2025            x1.01
```

r1.low step 2: likelihood

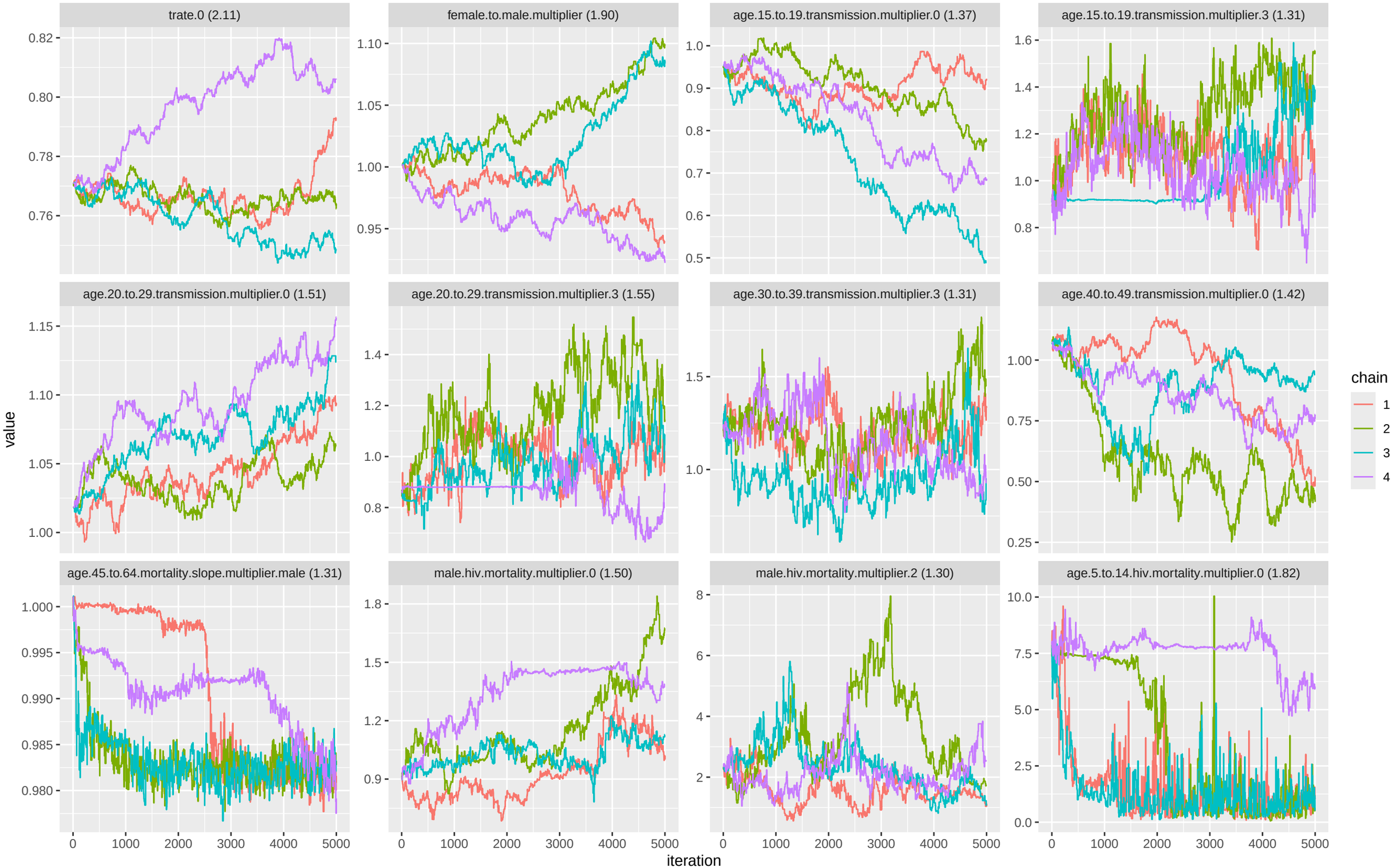
0
−10000
−20000
−30000
lue

0
25000
50000
75000
100000
iteration

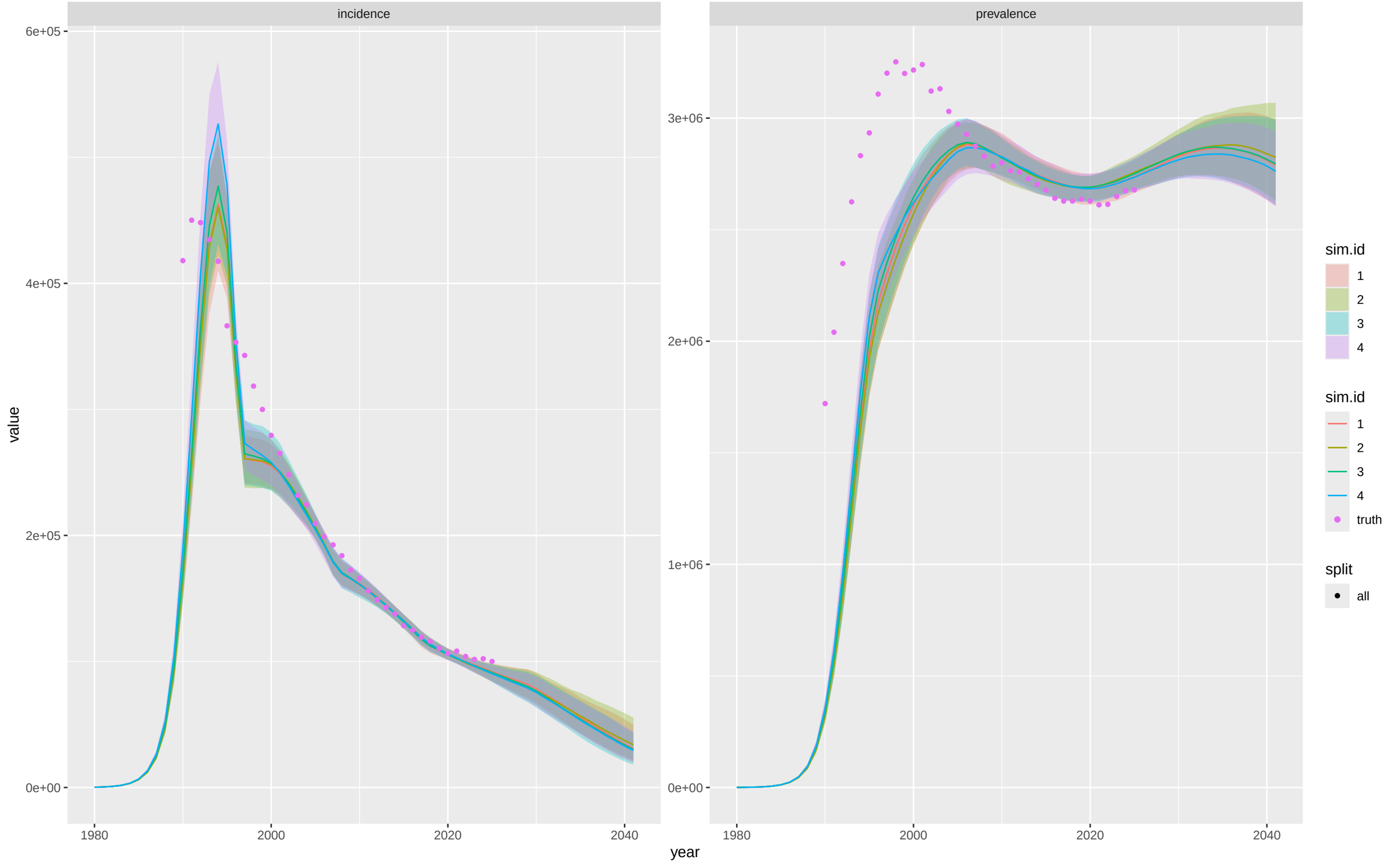
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

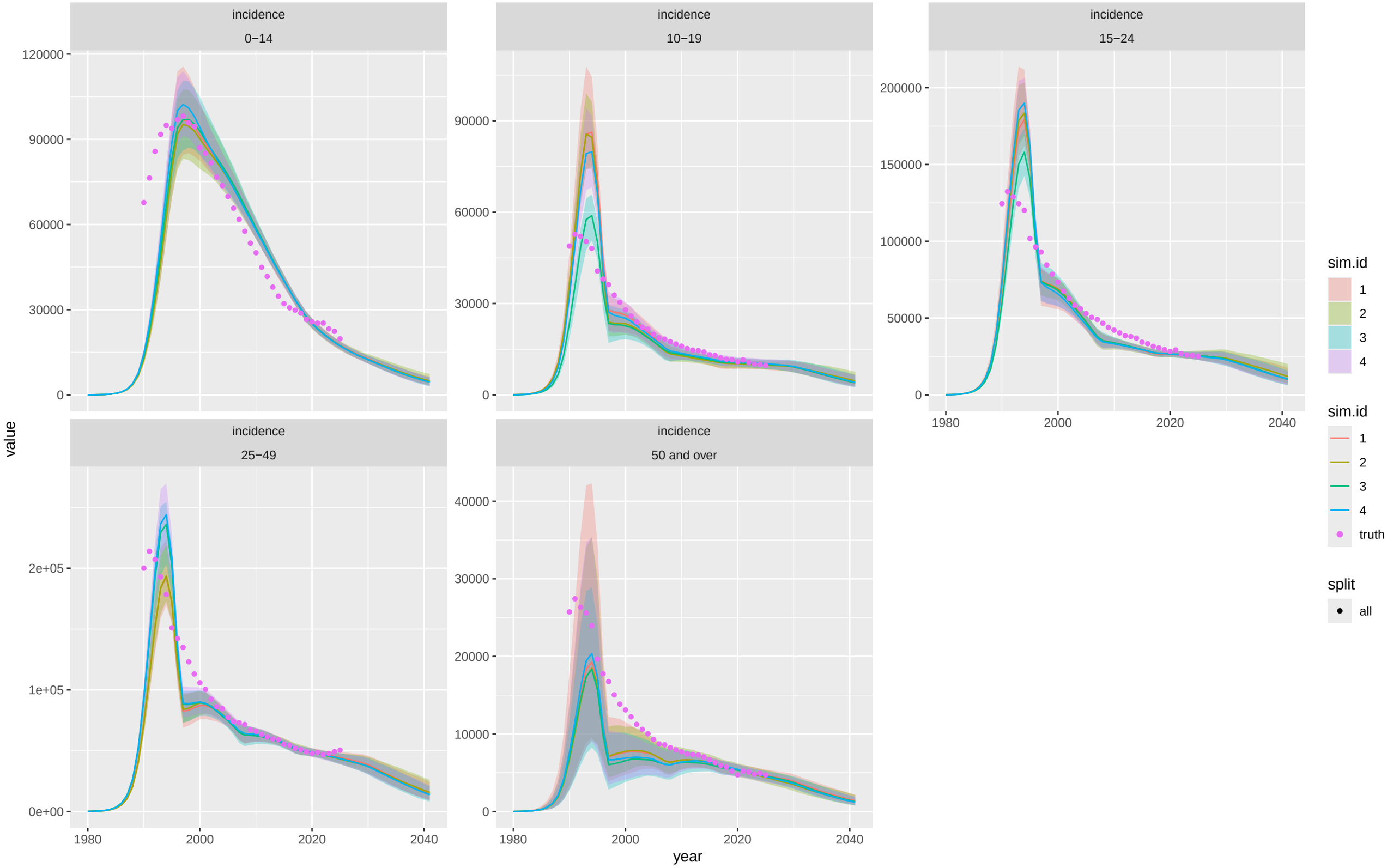
r1.low step 2: trace



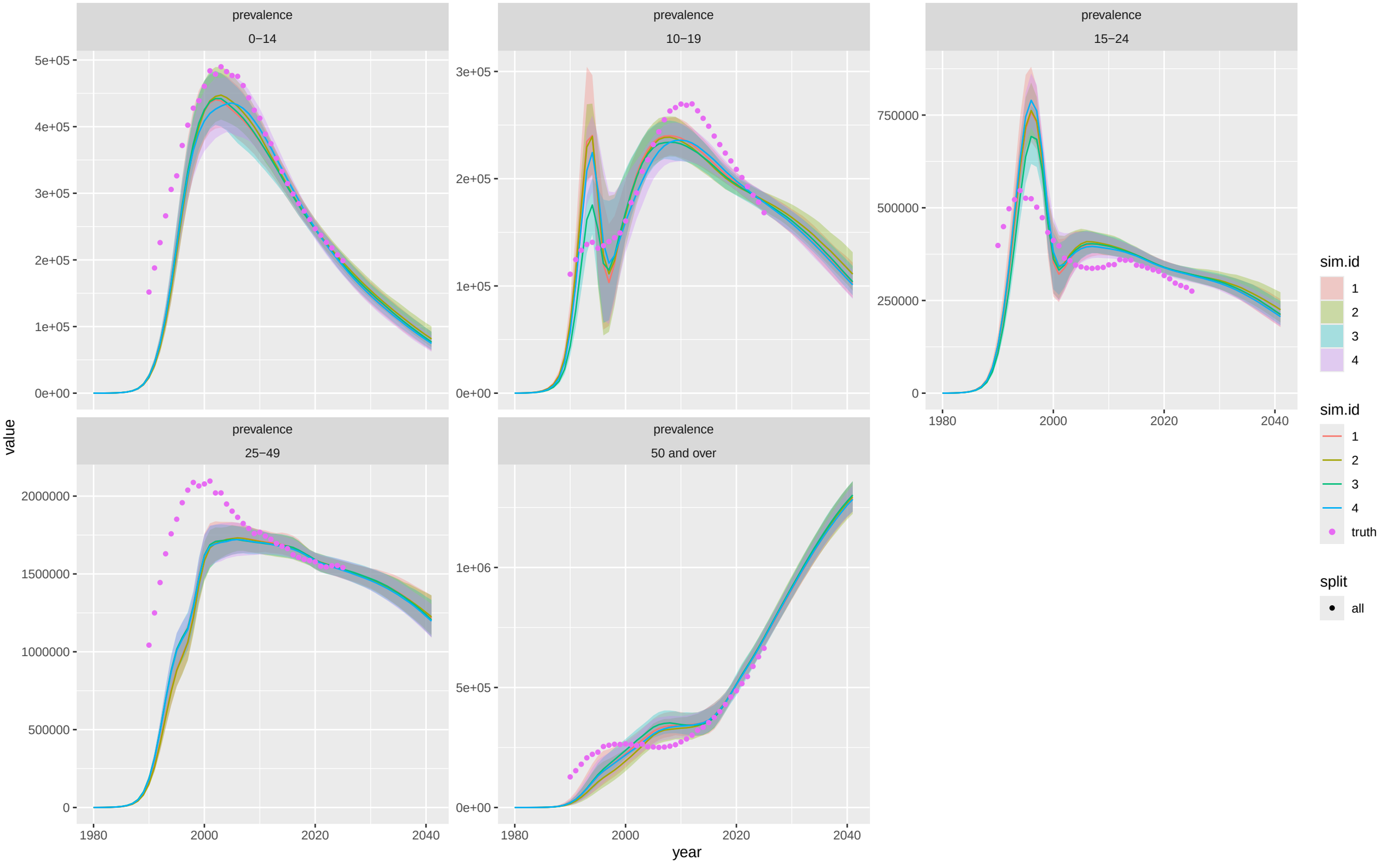
r1.low step 2: fit (one color per chain)



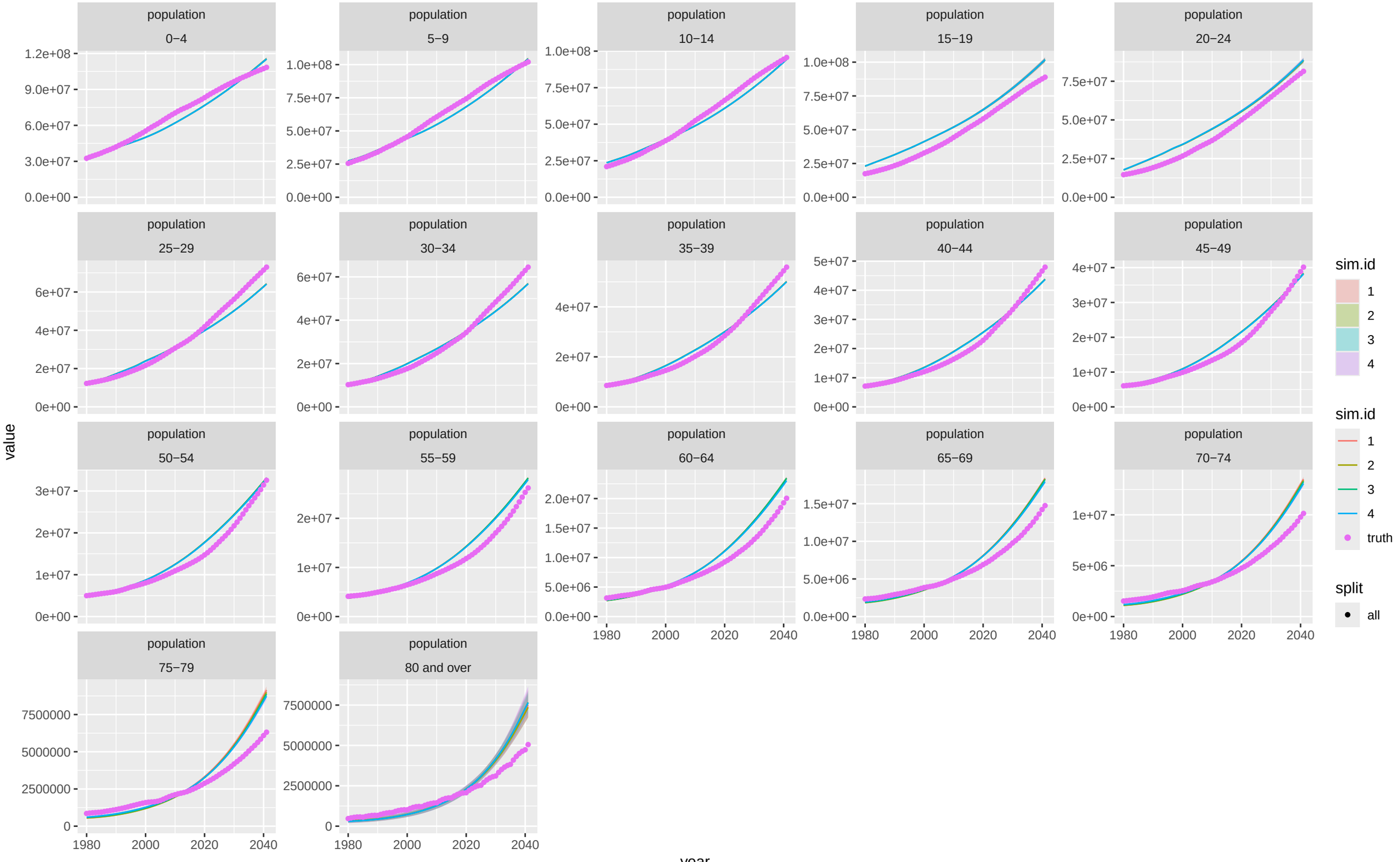
r1.low step 2: incidence.by.age (one color per chain)



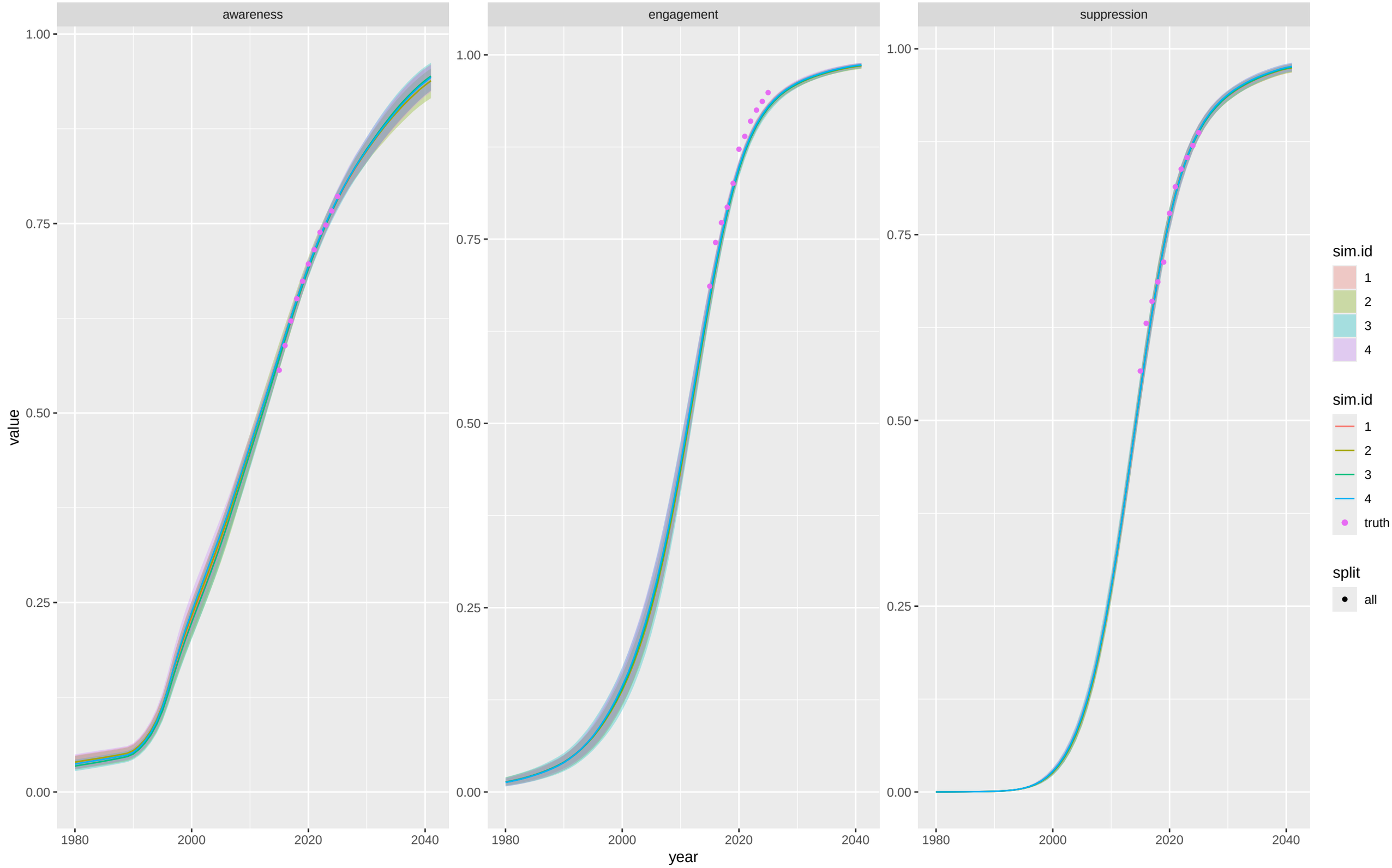
r1.low step 2: prevalence.by.age (one color per chain)



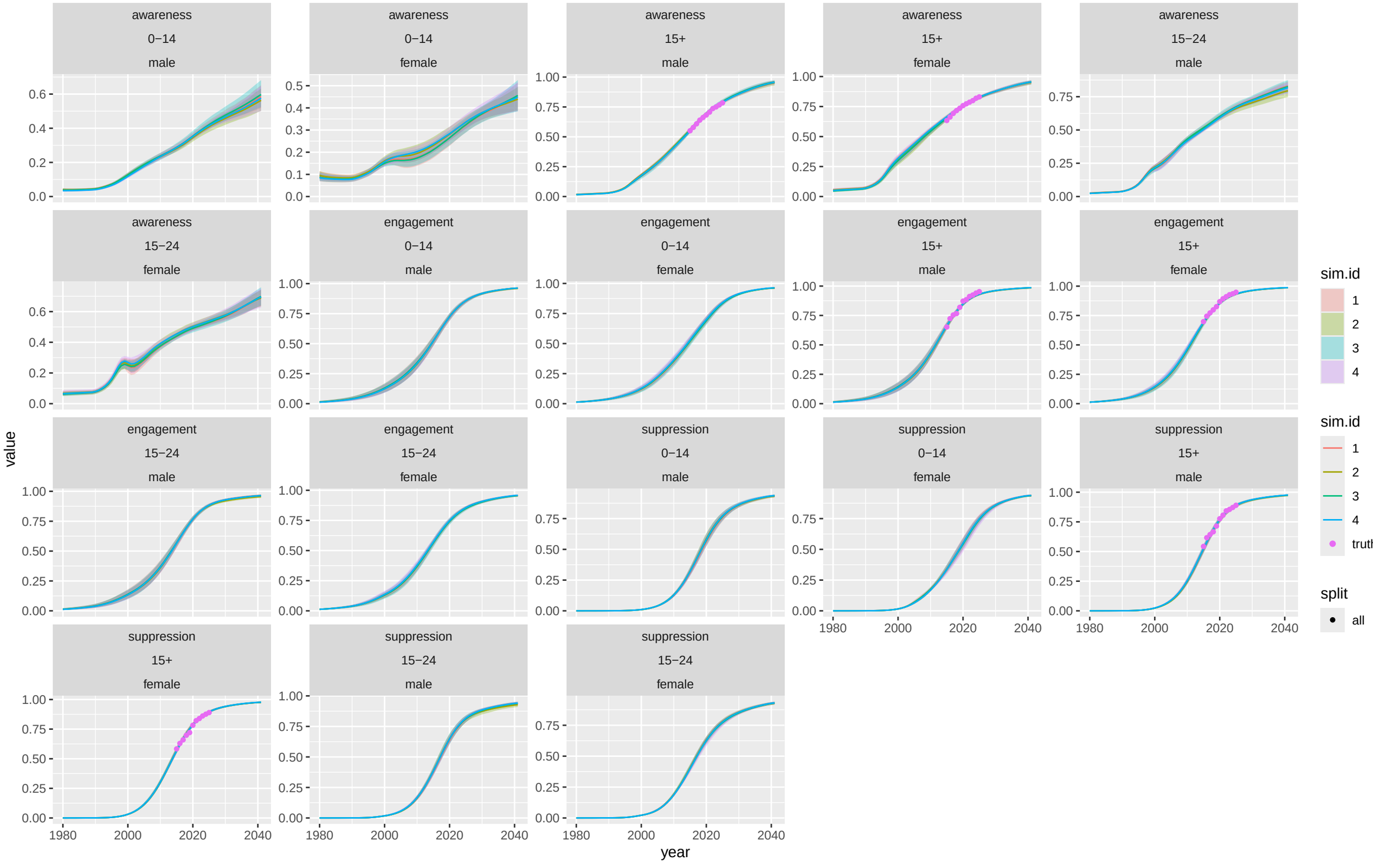
r1.low step 2: population.by.age (one color per chain)



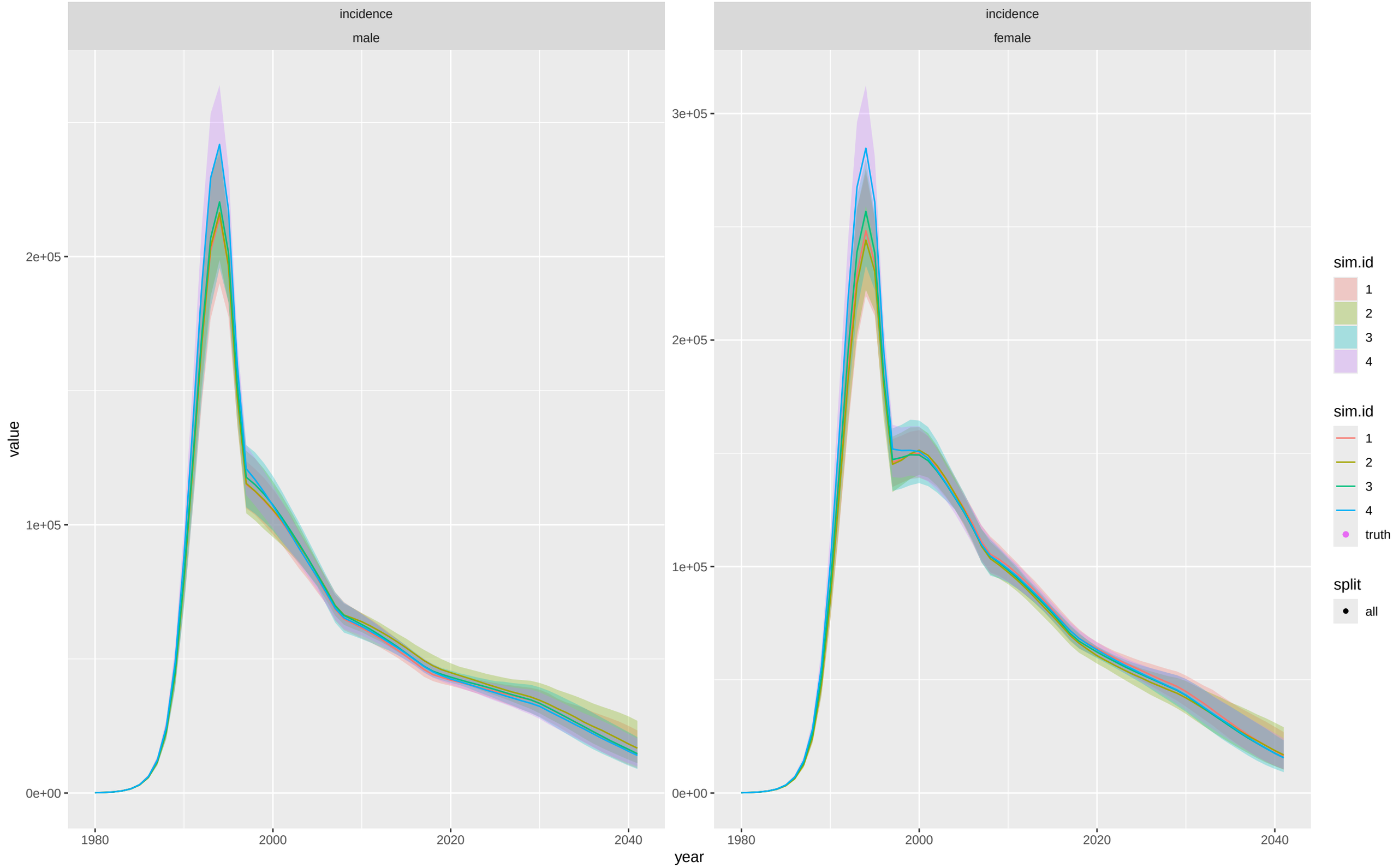
r1.low step 2: cascade (one color per chain)



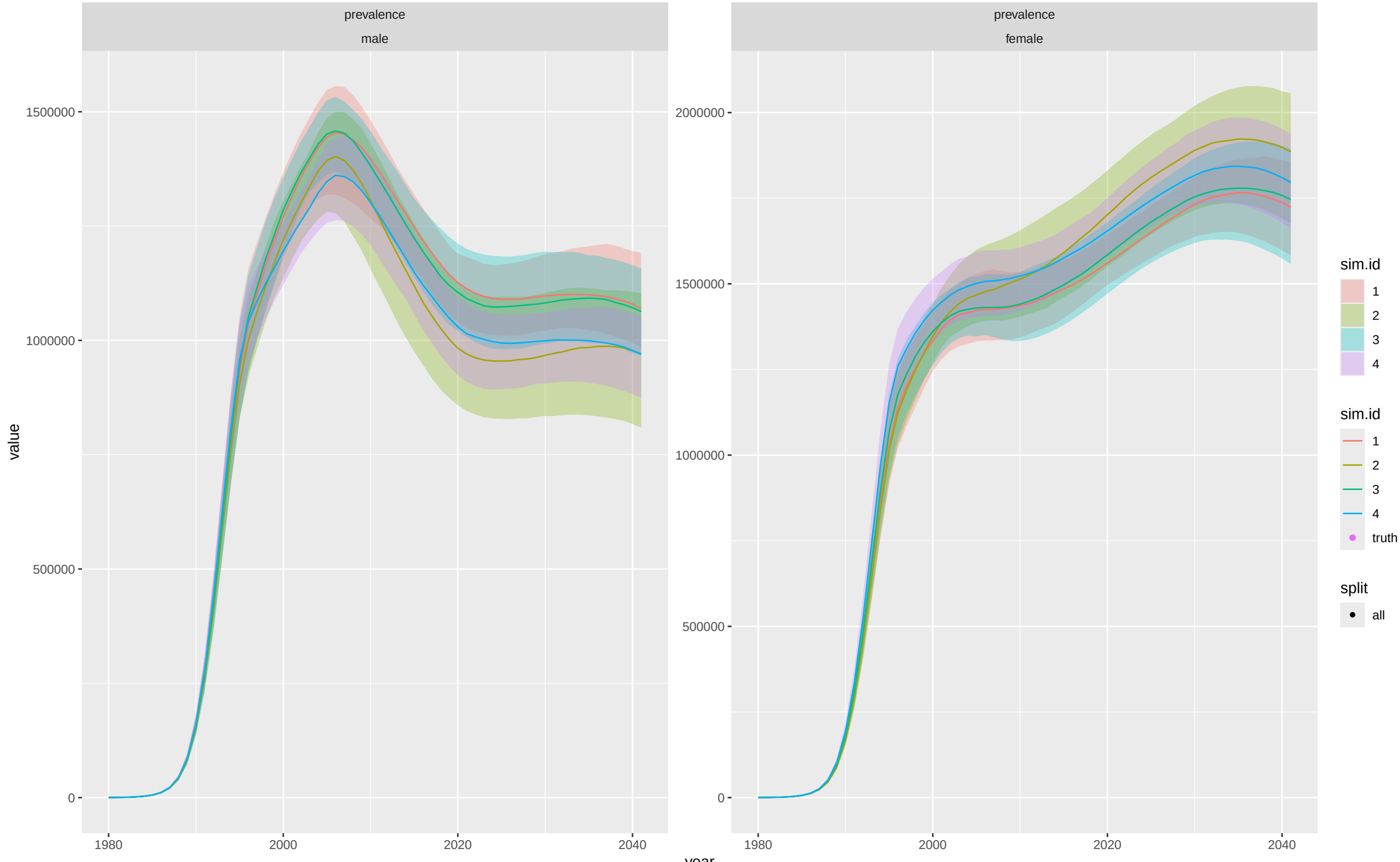
r1.low step 2: cascade.by.age.sex (one color per chain)



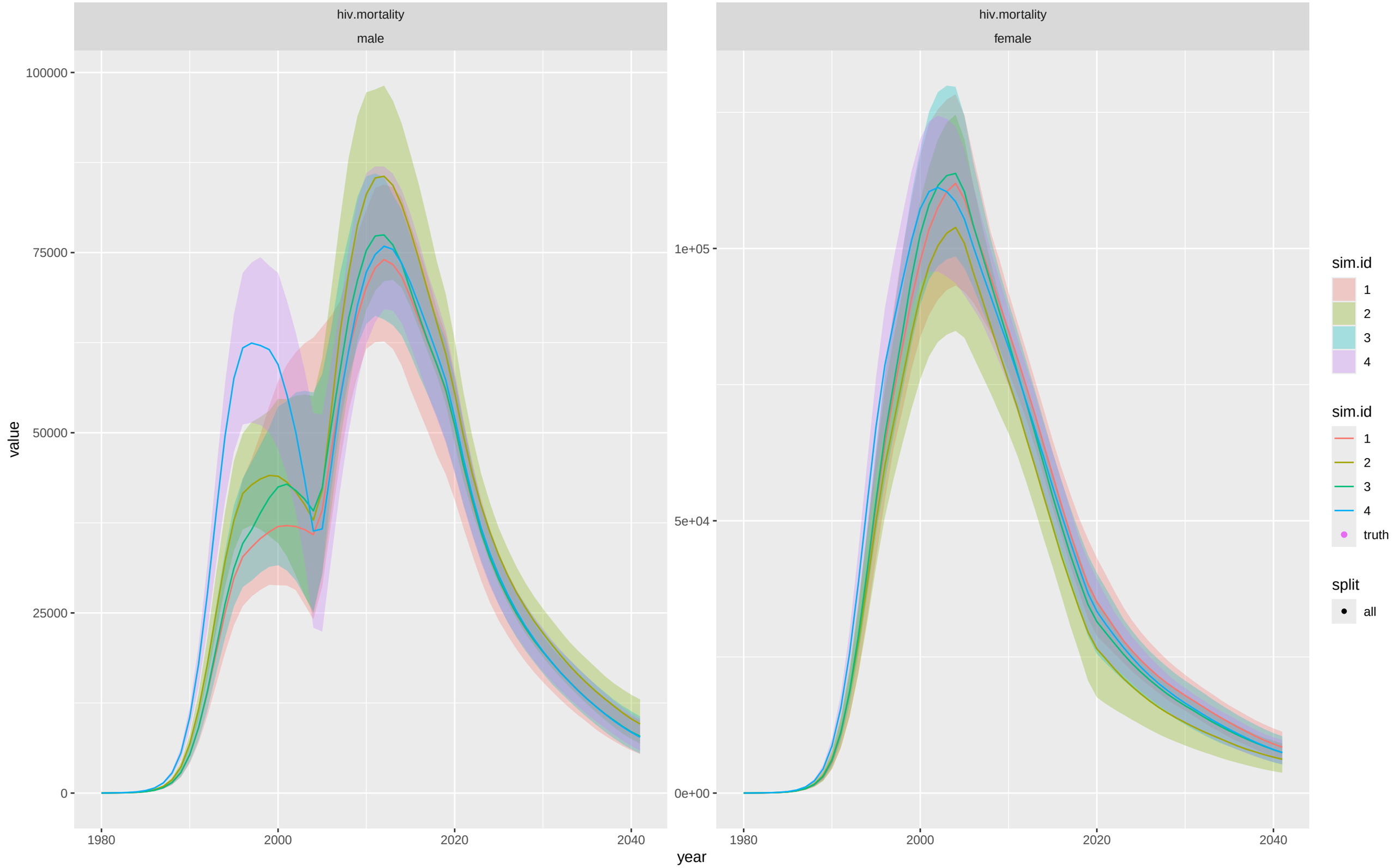
r1.low step 2: incidence.by.sex (one color per chain)



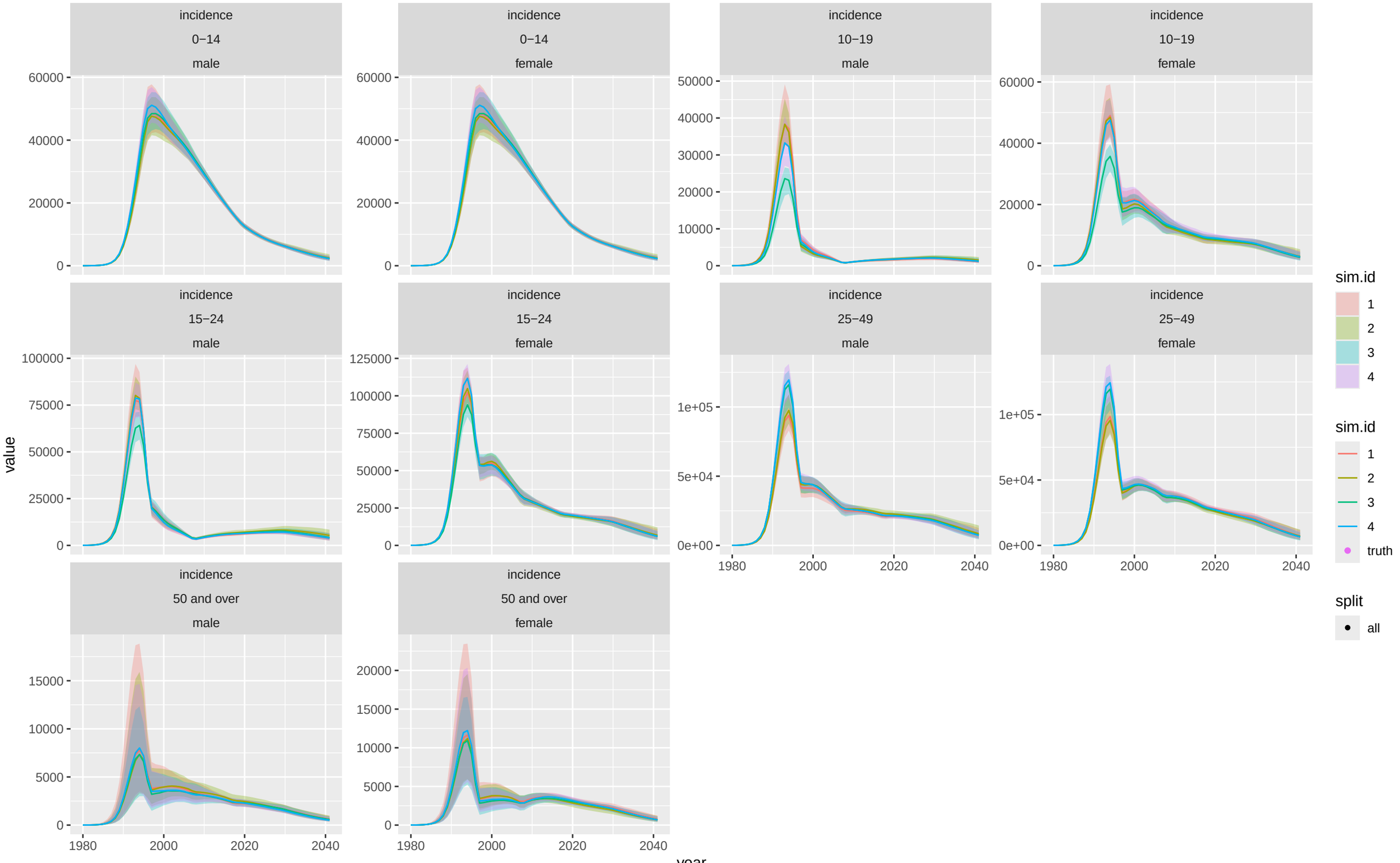
r1.low step 2: prevalence.by.sex (one color per chain)



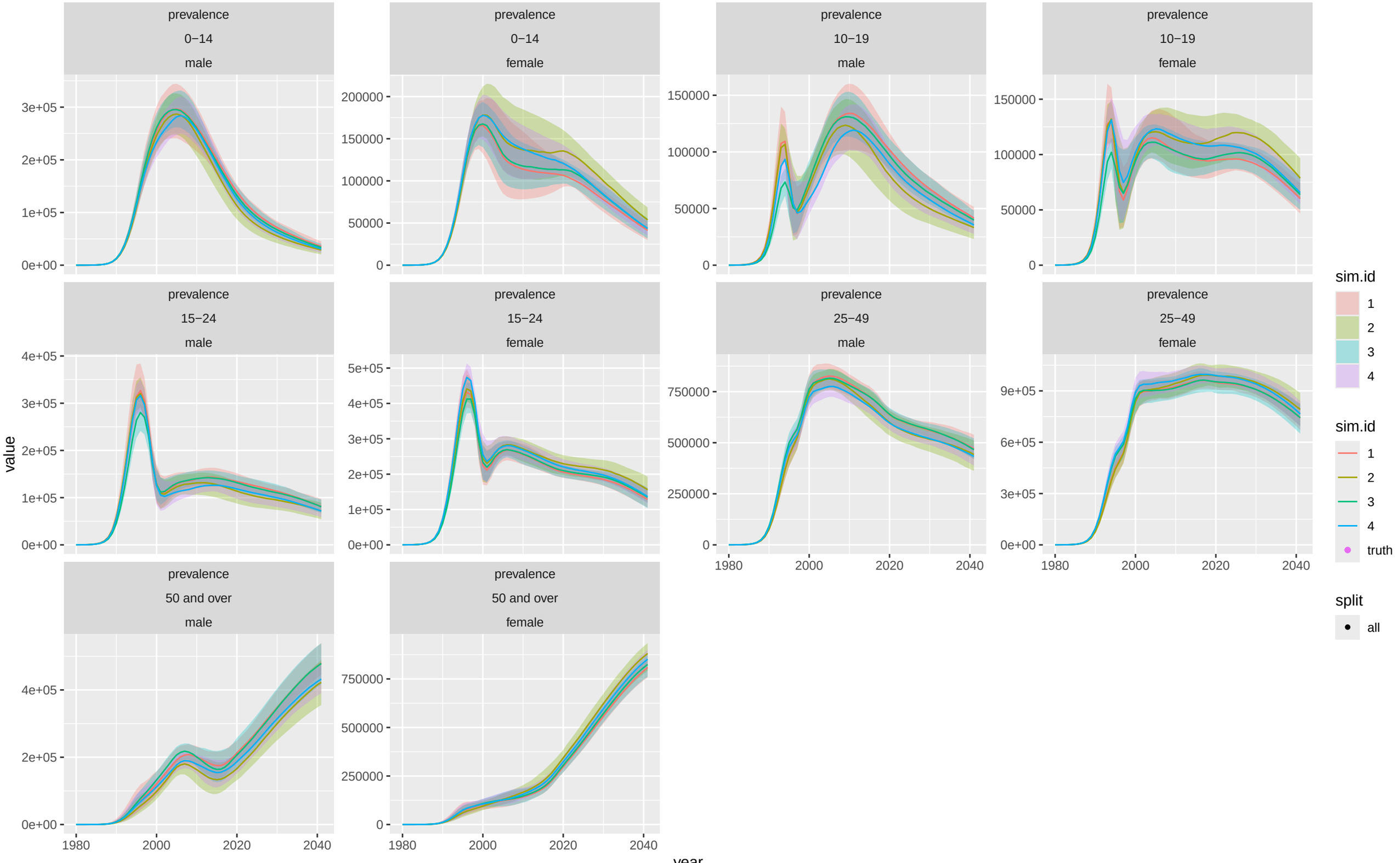
r1.low step 2: hiv.mortality.by.sex (one color per chain)



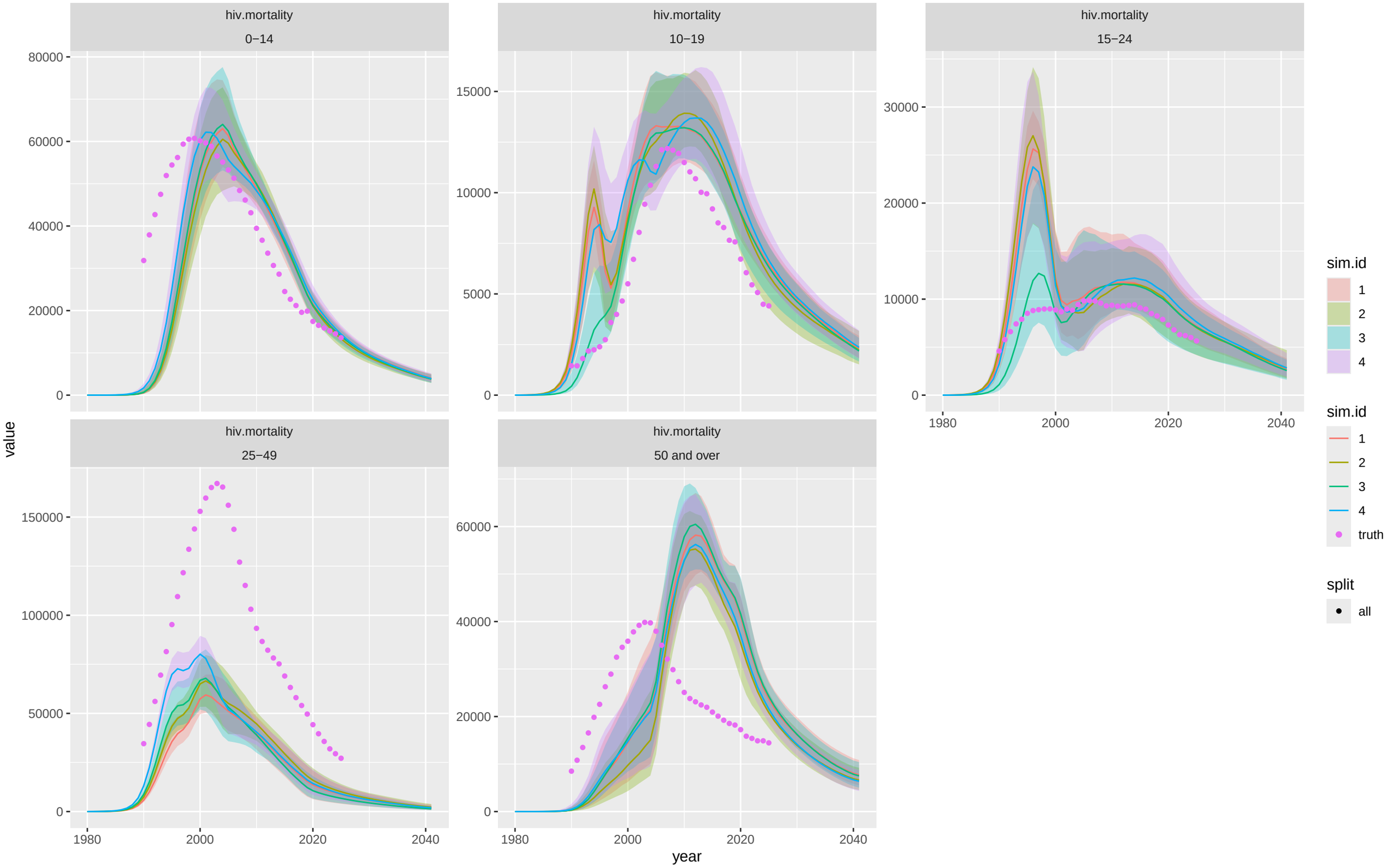
r1.low step 2: incidence.by.age.sex (one color per chain)



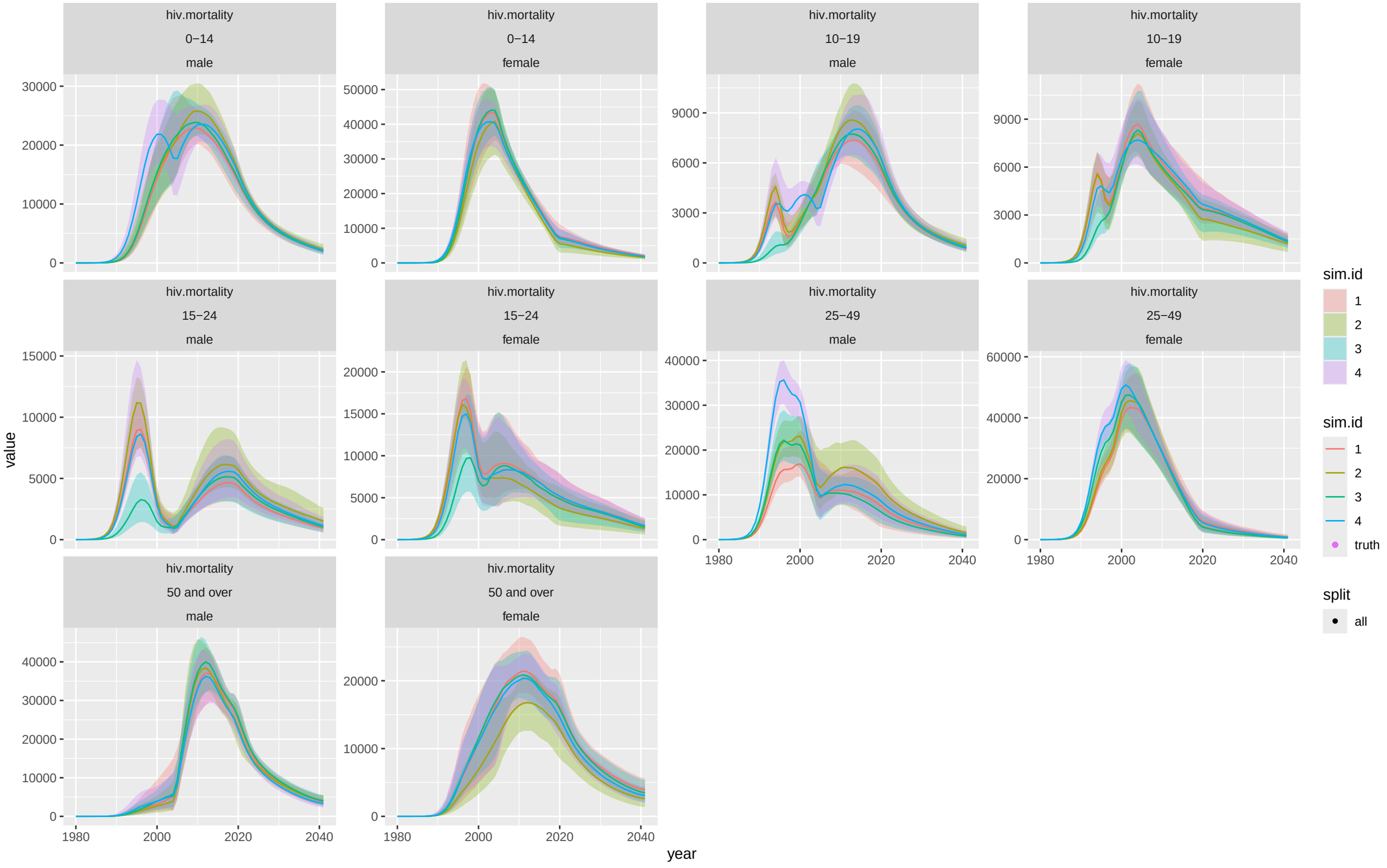
r1.low step 2: prevalence.by.age.sex (one color per chain)



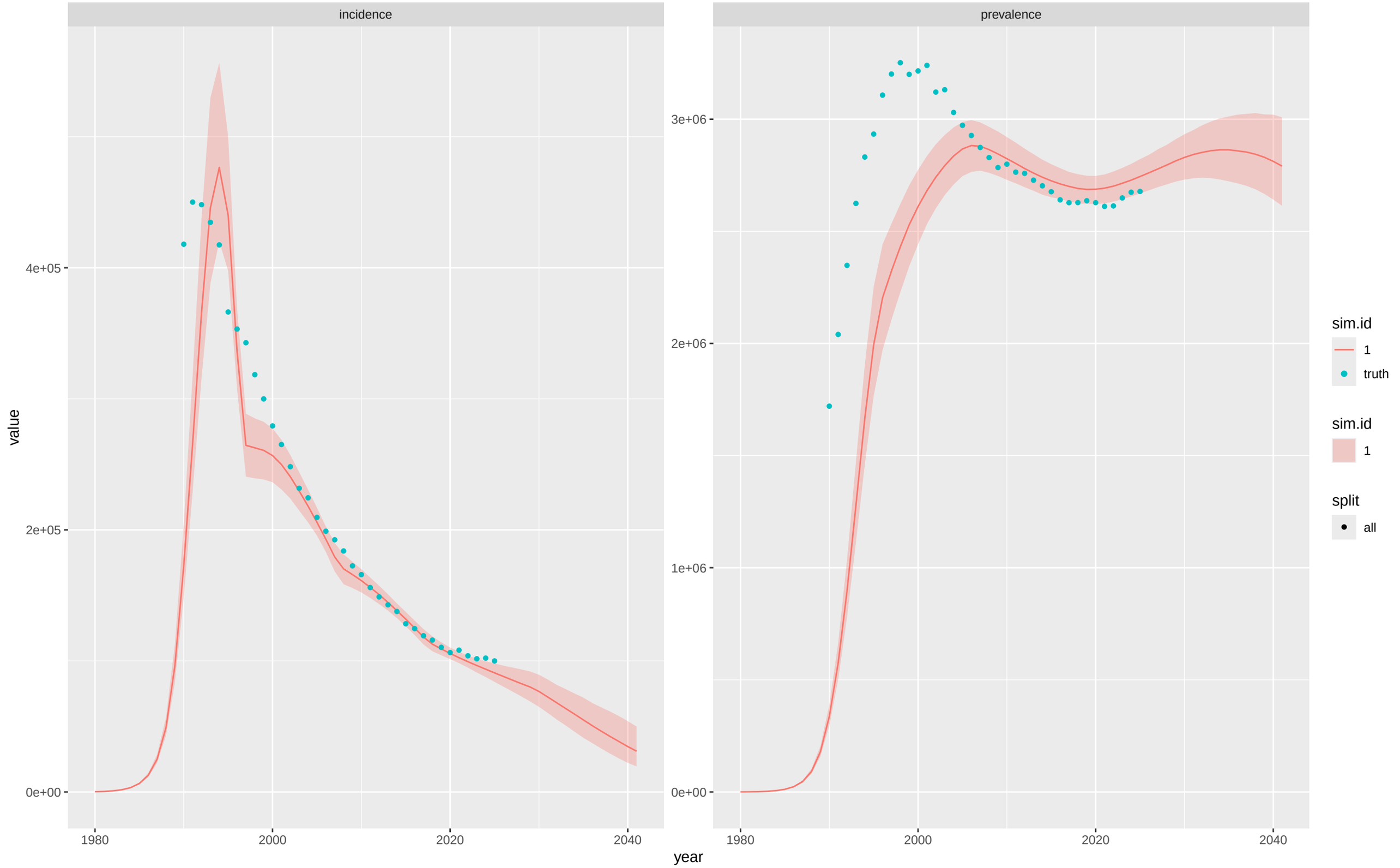
r1.low step 2: hiv.mortality.by.age (one color per chain)



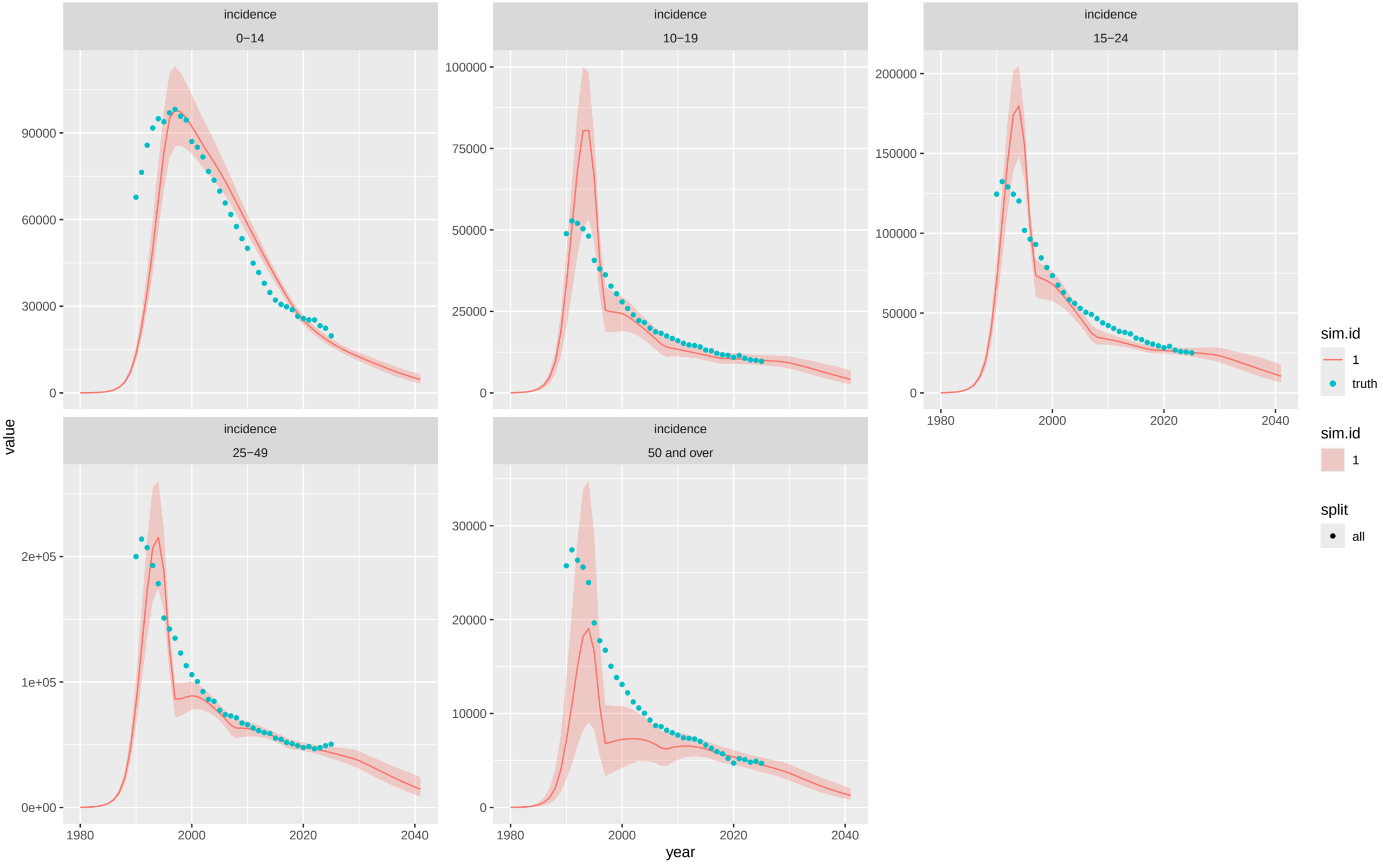
r1.low step 2: hiv.mortality.by.age.sex (one color per chain)



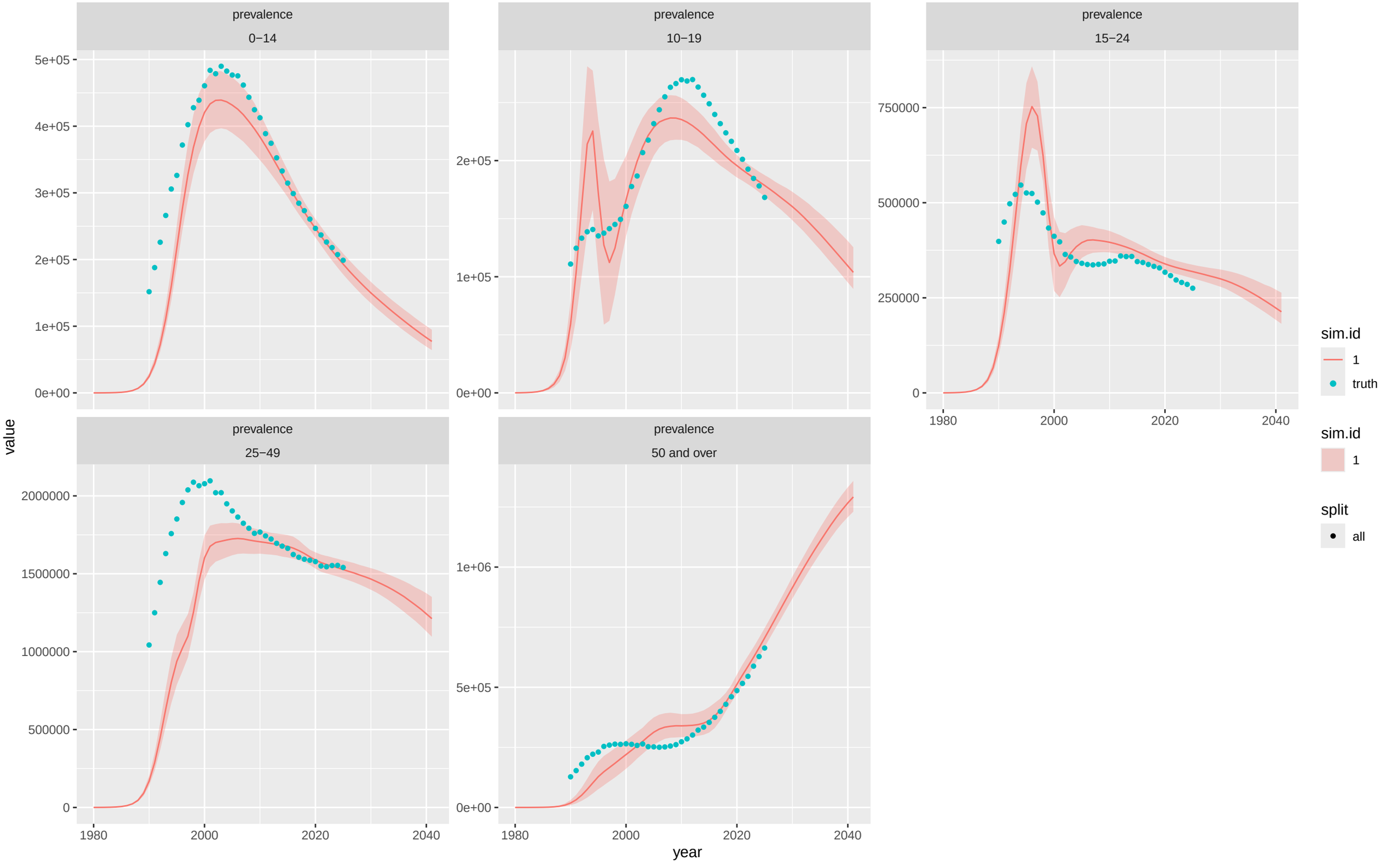
r1.low: fit (merged Step 2 simset)



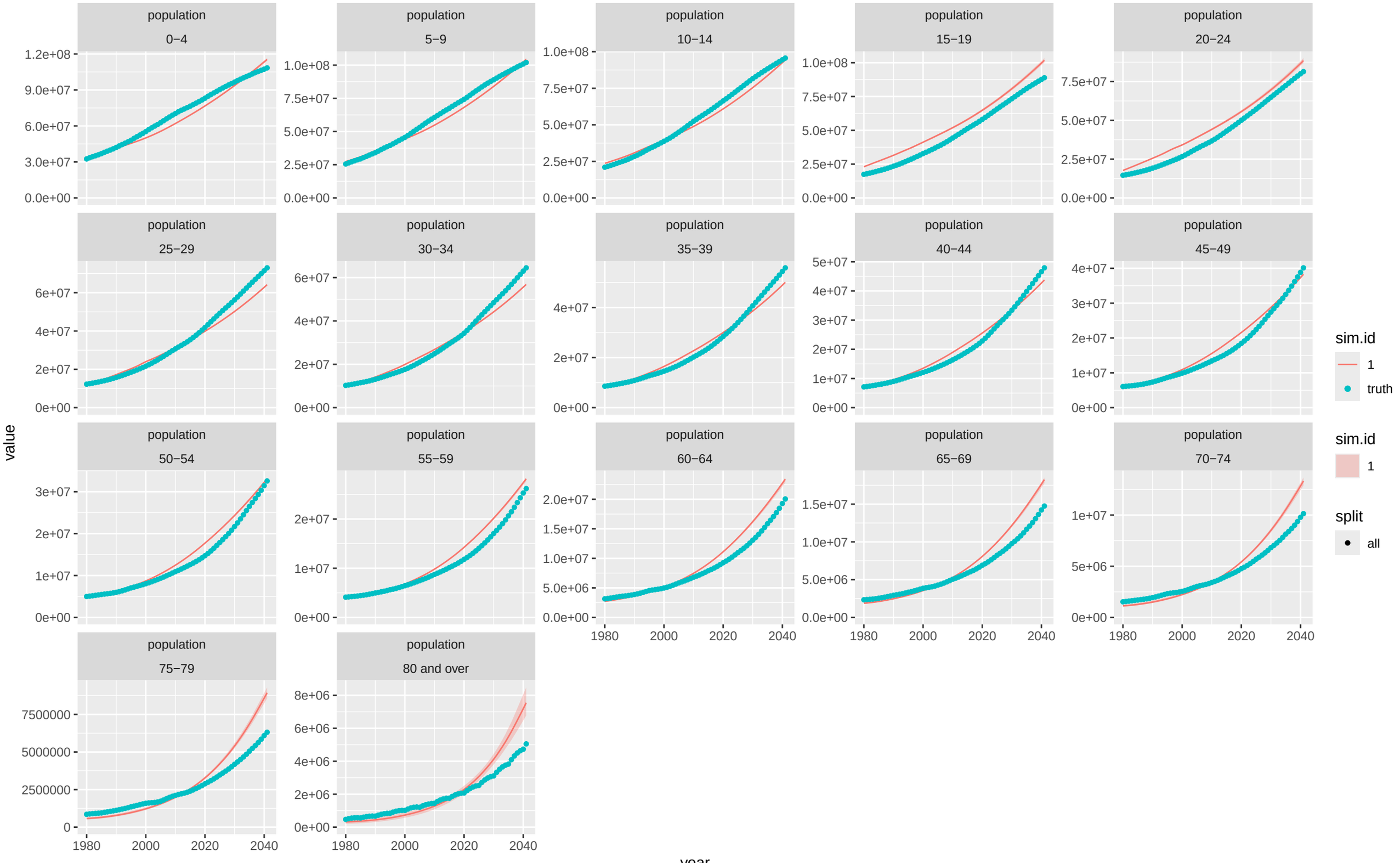
r1.low: incidence.by.age (merged Step 2 simset)



r1.low: prevalence.by.age (merged Step 2 simset)



r1.low: population.by.age (merged Step 2 simset)



r1.low: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

sim.id

1

split

all

r1.low: cascade.by.age.sex (merged Step 2 simset)

awareness 0−14 male
awareness 0−14 female
awareness 15+ male
awareness 15+ female
awareness 15−24 male
awareness 15−24 female
engagement 0−14 male
engagement 0−14 female
engagement 15+ male
engagement 15+ female
engagement 15−24 male
engagement 15−24 female
suppression 0−14 male
suppression 0−14 female
suppression 15+ male
suppression 15+ female
suppression 15−24 male
suppression 15−24 female

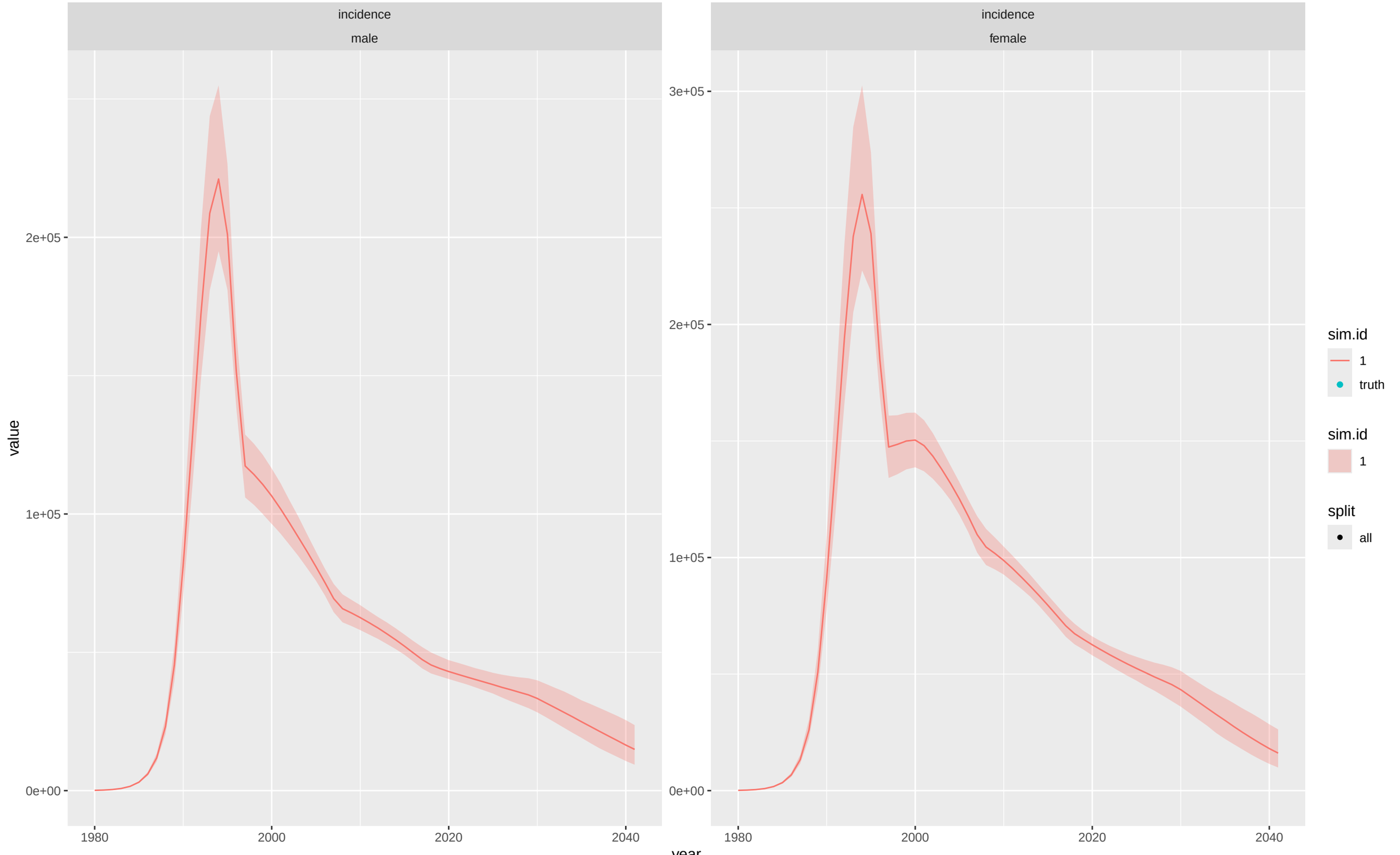
value
year

sim.id
1
truth

sim.id
1

split
all

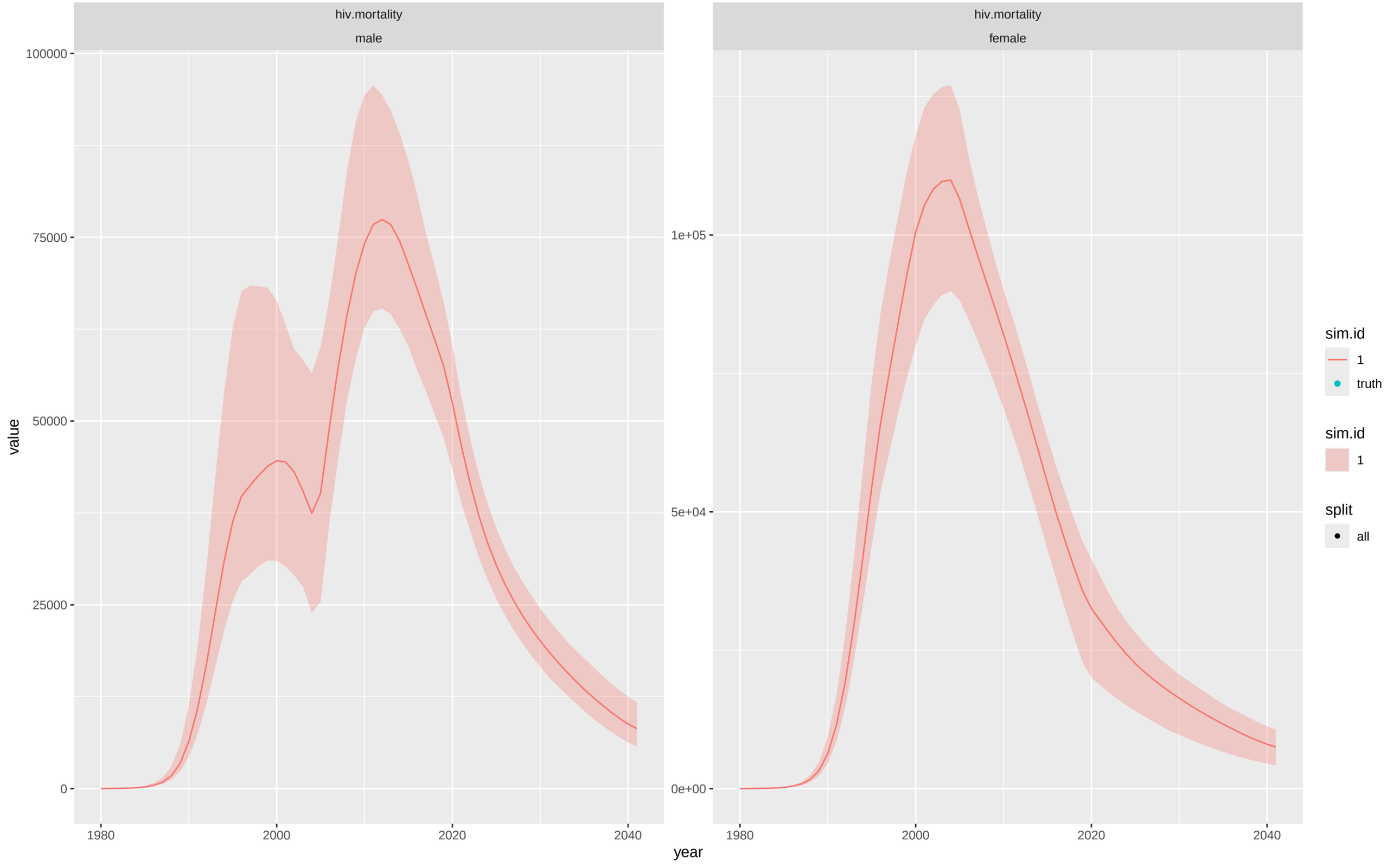
r1.low: incidence.by.sex (merged Step 2 simset)



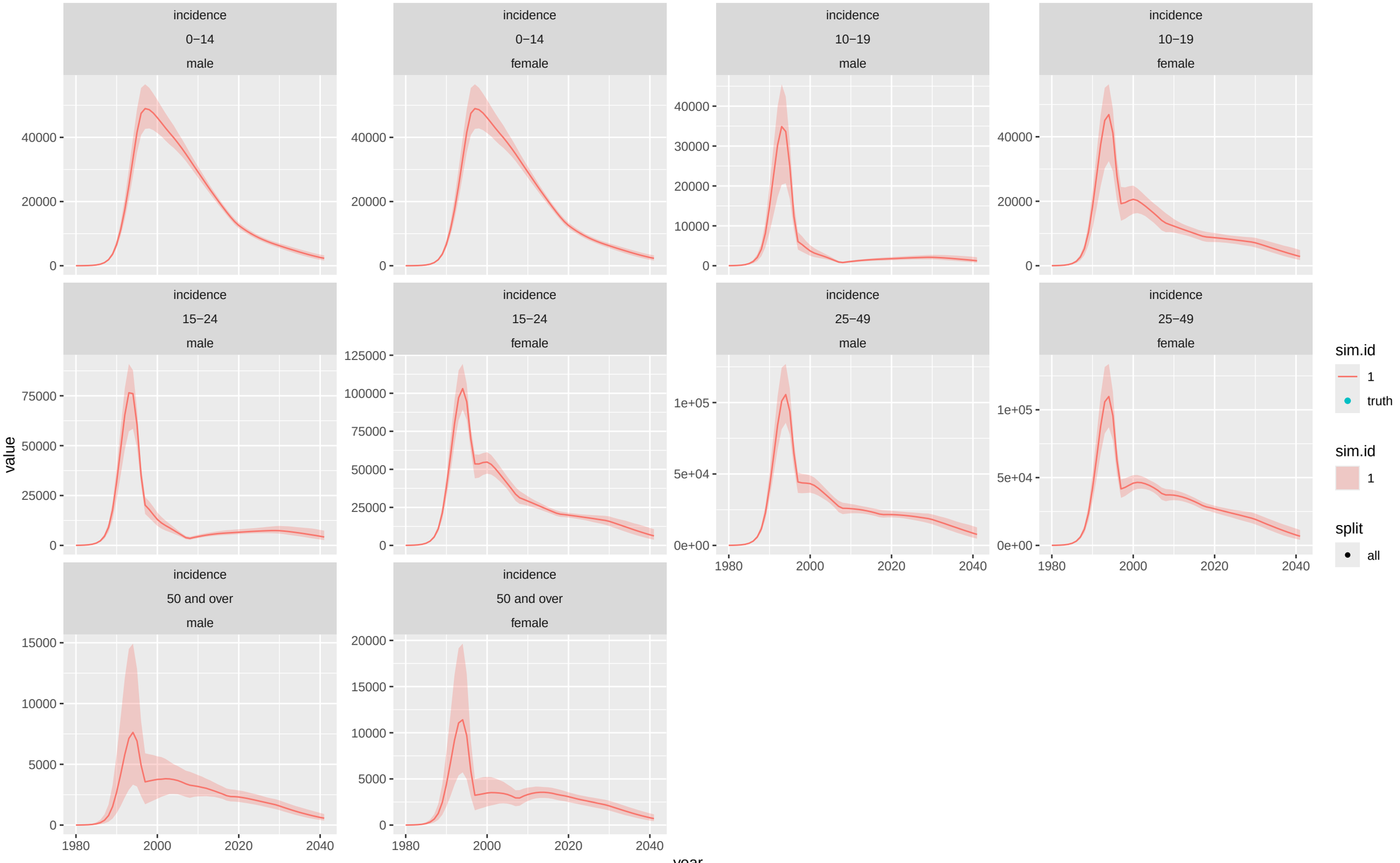
r1.low: prevalence.by.sex (merged Step 2 simset)



r1.low: hiv.mortality.by.sex (merged Step 2 simset)



r1.low: incidence.by.age.sex (merged Step 2 simset)



r1.low: prevalence.by.age.sex (merged Step 2 simset)

prevalence 0–14 male

prevalence 0–14 female

prevalence 10–19 male

prevalence 10–19 female

prevalence 15–24 male

prevalence 15–24 female

prevalence 25–49 male

prevalence 25–49 female

prevalence 50 and over male

prevalence 50 and over female

value

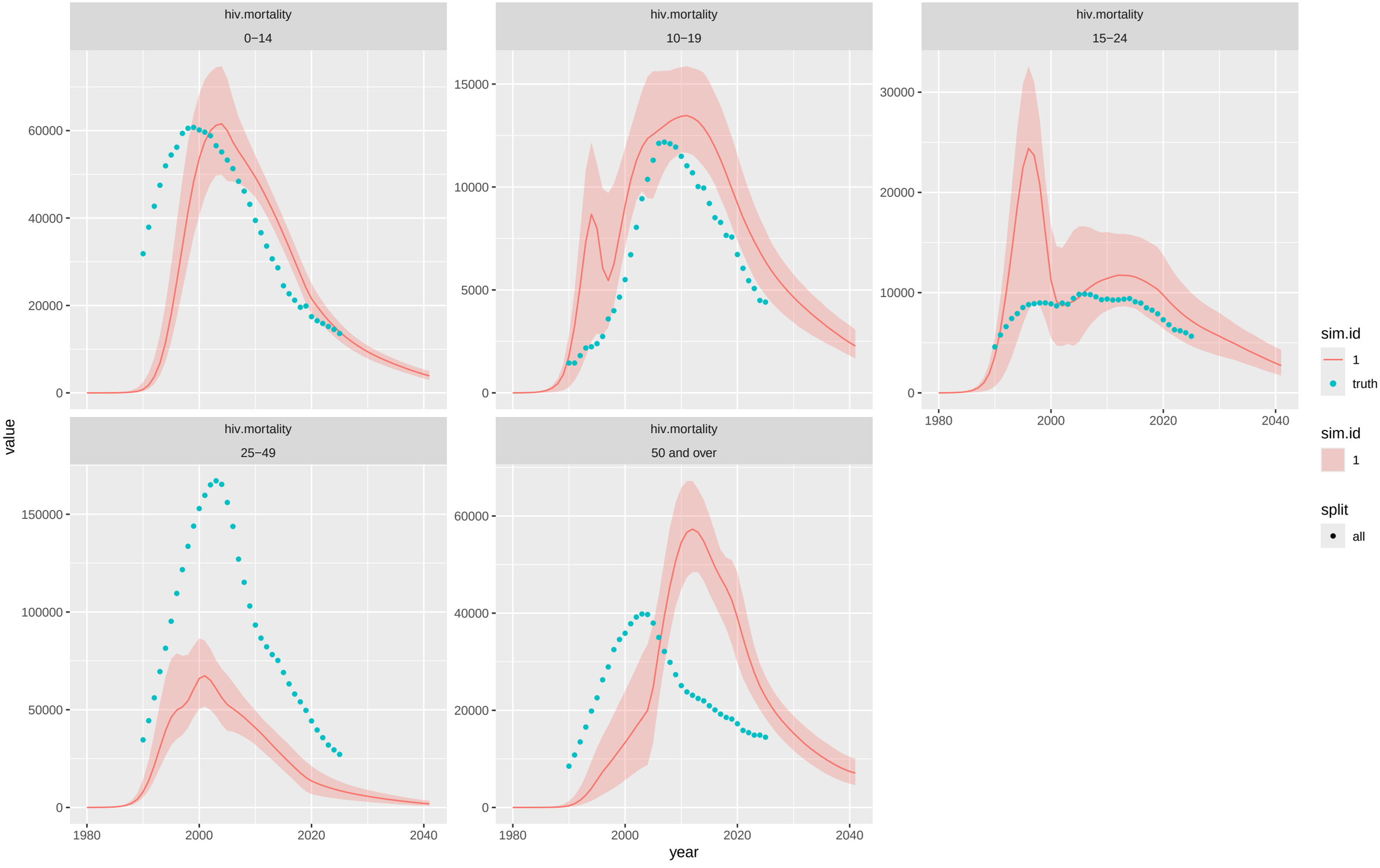
year

sim.id
1
truth

sim.id
1

split
all

r1.low: hiv.mortality.by.age (merged Step 2 simset)



r1.low: hiv.mortality.by.age.sex (merged Step 2 simset)

